A small JEPA world model knows what happened, not what will

Ravindu Somawansa

Abstract

We trained a LeWM-style JEPA world model (W1) on 182 teleoperated demonstrations of one SO-101 tabletop task for $3.30 and ran it through 20 pre-registered uses taken from published world-model papers. Its encoder knows what happened: on 69 policy rollouts it never saw in training, a linear probe reads the outcome of a finished episode at AUROC 0.9545, level with a hand-built colour judge (0.9432) and above a frozen DINOv2-S (0.8655), the commanded joints (0.7573) and policy identity (0.5882); at the grasp anchor the current latent reads the grasp outcome at 0.8199 (DINOv2-S 0.7067, joints 0.5305). It does not know what will happen: the imagined latent reads the grasp worse than the current one (0.70 vs 0.82), and ranking, progress, surprise monitoring, risk and multi-scale planning all miss their bars. One rig, one model, one data scale.

1. Introduction

World-model papers report the use that worked. We report every use we pre-registered for one small JEPA world model on one $200 arm: 20 gates, each with a bar, a dataset and a verdict, logged in a public register before or with the number (13 bars committed before their result, 6 in the same commit). 3 pass, 1 is met as predicted, 2 are not proven and 14 fail or are no-go. The result splits cleanly along the model. The *encoder* carries the episode's outcome, better than a frozen foundation encoder or the robot's own joint commands. The *predictor*, which every forecasting use relies on, does not turn that into a forecast. Contribution: (i) the register, with a load audit of every checkpoint use; (ii) controls that separate encoder from predictor; (iii) the reading that, at this data scale, the predictor is the weak link. **Caveats stated up front.** The encoder readouts are linear probes fit with leave-one-session-out cross-validation on 69 rollouts (44 success / 25 failure, 13 sessions) with operator labels; W1 was trained on teleop only and never saw them, but the controls were chosen after the data and are **exploratory**. Two of the three passes have same-commit bars.

2. Setup

Robot and data. One SO-101 arm, two cameras (top, wrist), one task family (put a bowl in another). W1 is trained on `ravediamond/so101_bowls_all_trimmed` (157 train / 25 validation episodes, split by grid layout); the evaluation rollouts are `ravediamond/so101-bowls-policy-rollouts`, a separate dataset of SmolVLA rollouts recorded after W1 was trained. **Model.** W1 follows the LeWM recipe (multi-view ViT encoder, action-conditioned predictor, SIGReg), trained on one rented GPU. **Controls.** W2′: frozen DINOv2-S, mean-pooled, top + wrist. Joints: the commanded joint vector. G9c-3: a hand-built colour judge on the final frame (no learning). Policy identity: which checkpoint ran the episode. **Load audit.** One early gate (G11-A) loaded W1 with `strict=False` and ran a random encoder. We therefore re-traced the load path of every use: 26 trained, 1 random (G11-A, already voided), 0 unknown. The VLA-JEPA world-model block loads 351 / 351 keys (0 missing).

3. The register

Table 1 lists the 20 uses, grouped by what they need from the model. *Encoder reads.* G10 step 1 reads the outcome at rest (pass, retrospective, same-commit); W1-C1's current-latent arm reads the grasp at the anchor (0.82); the pose probes (LeJEPA) give W1 a mean R^2 of 0.565 against W2′'s 0.867, so the encoder is neither mainly kinematics nor the better pose reader. *Predictor forecasts.* G0's open-loop prediction beats copy-last (0.31 vs 0.89 at K = 30): the predictor predicts its own latent. Every use of that prediction fails: G1b ranks chunks at top-1 0.60 (bar 0.8) while the commanded actions alone reach 0.68–0.90; W1-A's surprise catches 2/8 at 2/42; W1-C1's imagined latent (0.70) is below the current one; SURVEYOR's τ is 1.2488 (paper 0.20); HWM finds no crossover; G6a, G6b, W1-B, W1-C2, W1-R, REAP and G11-B-cal all miss their bars.

4. What the encoder knows

Table 2: outcome AUROC on the same 69 rollouts, leave-one-session-out, 95 % episode-bootstrap CIs. At rest W1 is level with the colour judge (CIs overlap) and above the frozen encoder and the joints; at the grasp anchor, where no judge applies, it leads both controls. The original G10 gate scored W1 at 1.000 on a larger rollout set; the rerun here, on the 69 Orin rollouts with the controls beside it, gives 0.9545.

feature	at rest	grasp anchor (65 scored, 21 fail)
W1 (ours)	0.9545 [0.9023, 0.9908]	0.8199
colour judge G9c-3 (no learning)	0.9432 [0.8919, 0.9878]	—
W2′ frozen DINOv2-S	0.8655 [0.7767, 0.9405]	0.7067
commanded joints	0.7573 [0.6084, 0.8880]	0.5305
policy identity	0.5882 [0.4084, 0.7515]	—

Table 1: Outcome AUROC, same rollouts, LOSO probes (exploratory).

5. Where the predictor fails

The cleanest contrast is W1-C1: same encoder, same anchor, current latent 0.82 vs imagined 0.70 (bar ≥ 0.85, $\geq +0.05$ over current). Imagining forward loses outcome information the encoder had. In SURVEYOR, W1's latent-distance threshold fails both checks, while W2′ (τ 0.3601) and the joints (τ 0.0265) pass the scale check S2: the predictor-facing geometry of W1 is the problem, not the task. *Objection: G10 step 2 is a predictive pass.* The imagined latent beat policy identity by +0.179 — but that bar is same-commit and retrospective, and the two bar-first reruns of the same idea (W1-C1, G11-B-cal) fail.

6. Limitations

Exploratory controls: 69 rollouts, operator labels, probes chosen after the data; fresh rollouts scored under a bar committed first are the test. One rig, one model, one data scale: a second scale point (653 episodes) is costed and queued separately. Not run: W1-A surprise and W1-C1 imagined with W2′ / joints (each needs its own predictor), G6a T1 with the controls (needs a reimplementation). G0 and G10 bars are same-commit.

7. Conclusion

On a small teleop dataset, a JEPA world model learns an encoder that knows how the episode went and a predictor that cannot say how it will go. Reporting every pre-registered use, not the best one, is what makes that visible.